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ORIGINAL ARTICLE

Progressive natural killer cell dysfunction in advanced-stage clear-cell renal cell carcinoma and association with clinical outcomes

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Background: Natural killer (NK) cells are important contributors to antitumor immunity in clear-cell renal cell carcinoma (ccRCC). However, their phenotype, function, and association with clinical outcomes in ccRCC remain poorly understood.

Materials and methods: We analyzed single-cell RNA sequencing data from 13 primary tumors, 1 localized tumor extension, and 1 metastasis from ccRCC patients at different clinical stages. For each primary tumor specimen, paired normal kidneys were also analyzed. Differential gene expression analysis was carried out to investigate NK cell phenotypes and to derive a gene expression signature. Gene signatures from NK cell subclusters of interest were used to interrogate bulk transcriptomic datasets and expression with clinical outcomes. Finally, tumor-infiltrating NK cell function (cytokine production and cytotoxicity) was assessed by isolation of live NK cells from ccRCC tissue, co-culture with K562 target cells, and measurement of cytokine production (interferon-γ) and cytotoxicity (CD107a) markers by flow cytometry.

Results: Single-cell transcriptomic data were analyzed from 13 patients with ccRCC (tumor/normal kidney), resulting in 21 139 NK cells. Clustering analysis revealed six NK cell subsets. Bright-like NK cells were significantly enriched in advanced ccRCC compared with localized ccRCC and normal kidney, expressed markers of tissue residency (*ZNF683*/Hobit, *ITGA1*/CD49a, *CD9*, *ITGAE*/CD103), and had decreased expression of cytotoxicity genes (*GZMB*/Granzyme-B, *PRF1*/perforin). In independent cohorts (The Cancer Genome Atlas ccRCC cohort, CheckMate 025), a gene expression score representing this dysfunctional NK cell phenotype was enriched in advanced ccRCC and was associated with worse overall survival. Functional interrogation of tumor-infiltrating NK cells from ccRCC confirmed that tumor-resident CD49a+CD9+ NK cells had impaired cytotoxicity compared with CD49a−CD9− NK cells.

Conclusions: A dysfunctional, tumor-resident NK cell phenotype was enriched among patients with metastatic disease and associated with worse survival in patients with advanced ccRCC across multiple patient cohorts. Restoration of NK cell function (via cytokine stimulation or NK cell engineering) could provide a novel avenue for therapeutic intervention against ccRCC.

Key words: renal cell carcinoma, single-cell transcriptomics, immunogenomics, NK cells

INTRODUCTION

Initial successes in immunotherapy have revolutionized the treatment of clear-cell renal cell carcinoma (ccRCC).[1,2] Immune checkpoint inhibitors, both alone and in combination with vascular endothelial growth factor receptor-targeted therapies, are now the standard of care and have improved progression-free survival (PFS), overall survival (OS), and quality of life among patients with advanced or metastatic ccRCC.[1,2] Despite these significant advances, nearly all patients develop disease progression after treatment with checkpoint inhibitors, and novel immunologic strategies to address checkpoint inhibitor-refractory renal cell carcinoma

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(RCC) are needed. Recent work has also revealed that immune exhaustion across multiple cell types contributes to dysfunctional antitumor cytotoxicity and poor prognosis in ccRCC, and an improved understanding of immune cell subsets may reveal new targets for immunotherapy.[3,4]

Natural killer (NK) cells are thought to play a key role in the immune response against cancer, including ccRCC.[5] NK cell function is diverse and relies on a host of activating and regulatory immune receptors.[6] Killer immunoglobulin-like receptors and CD94/NKG2A expressed by NK cells are key inhibitory receptors that recognize human leukocyte antigen (HLA) class I molecules expressed by healthy cells, thereby permitting self-tolerance when in contact with healthy cells. This process requires periodic cross-talk between NK cell receptors and HLA class I to maintain self-tolerance, allowing a competent NK cell to recognize abnormal tumor cells that often reduce expression of HLA class I machinery.[7] NK cells have historically been classified according to their expression of CD56 and CD16, with $CD56^{bright}CD16^{low/-}$ NK cells having immunomodulatory and cytokine-producing roles and $CD56^{dim}CD16^{+}$ NK cells having a more cytotoxic role. However, recent advances in NK cell phenotyping with multiparameter cytometry and proteomic analyses have identified more nuanced functional subsets.[8] Additionally, among some tumor types, patients with advanced cancer NK cells display a dysfunctional phenotype marked by altered gene expression profiles and reduced cytotoxic function.[9] Recent work based on single-cell RNA sequencing (scRNA-seq) identified a population of infiltrating intraepithelial type 1 innate lymphoid cells (ILC1s) in ccRCC, which was associated with worse OS.[10] However, currently there is a paucity of data regarding the transcriptomic landscape and function of NK cells, particularly tumor-infiltrating subtypes, in ccRCC. We, therefore, carried out single-cell transcriptomic analysis of NK cells in 13 patients with ccRCC across different disease stages, identifying an enrichment of dysfunction bright-like NK cells in advanced ccRCC. We further validated this enrichment and observed an association with worse OS in two large, independent cohorts. Finally, we carry out *in vitro* functional analysis to validate the diminished cytotoxicity of this population in an additional cohort of five patients. Overall, this study identifies NK cell dysfunction as a key component of impaired antitumor immunity in advanced ccRCC.

MATERIALS AND METHODS

Study design and participants

The samples used in the single-cell analysis for this study are the same as those used in our prior analysis of T-cell and macrophage function in ccRCC.[3] These include primary kidney tumor and adjacent healthy tissue specimens from 13 patients obtained by nephrectomy as well as one adrenalectomy from a patient with locally advanced disease, and one abdominal mass resection from a patient with metastatic disease (Figure 1A). Sample acquisition and processing were previously described in detail and are summarized below.[3] We identified 13 patients with pathologically confirmed ccRCC who were treated at Dana-Farber Cancer Institute. Institutional review board approval (Dana-Farber Cancer Institute) and informed consent were obtained from all patients before tissue acquisition and analysis.

Data processing, filtration, and clustering

Raw count data and metadata of 164 722 cells that were previously analyzed were reconstructed into a Seurat object in R (GNU Project).[3] All processing steps were carried out using 'Seurat v5.0.1'.[11] Unless noted, default parameters were used in processing steps.

Normalization of data was carried out using the 'NormalizeData' function. We carried out principal component analysis (PCA) and kept the top 20 principal components (PCs) for batch correction using the Harmony algorithm (implemented using 'harmony v1.0.1') via the 'RunHarmony' function with parameters group.by.vars = 'Batch', reduction = 'pca', reduction.save = 'harmony'.[12] We constructed a shared nearest neighbor (SNN) graph using the top 20 PCs using 'FindNeighbors' (dims = 1 : 20, reduction = 'harmony'). The cells were clustered using the 'FindClusters' function using the Louvain community detection method and projected on to uniform manifold approximation and projection (UMAP) via the 'RunUMAP' function (reduction = 'harmony', dims = 1 : 20).

Then, we extracted previously identified NK cell types ('NK cell.1', 'NK cell.2', 'NK cell.3', and 'proliferating NK cell') from 'ClusterName_ImmuneCells' annotation in the metadata. The resulting NK cell data were filtered by keeping samples containing at least 45 NK cells.

We carried out PCA on the NK cells using the 'RunPCA' function. We then calculated an SNN graph by using the top 20 PCs using the 'FindNeighbors' function with the parameter dims = 1 : 20. The number of PCs was determined using 'ElbowPlot'. To identify groups of similar NK cells, we carried out clustering analysis based on the SNN using the Louvain community detection method ('FindClusters' with resolution = 0.5). For further analysis, we removed proliferating NK cells resulting in a total of 21 139 NK cells.

For visualization, we projected the clusters on to UMAP via the 'RunUMAP' function (reduction = 'harmony', dims = 1 : 20). The clustering analysis resulted in six distinct NK populations and was visualized using 'DimPlot' and 'DimPlot_scCustom' from 'scCustomize v1.1.3'.[13]

NK cell cluster identification

From six distinct clusters, the NK cell populations were annotated based on the average expression of canonical marker genes for bright genes and dim/$CD16^{+}$ genes in each cluster[14] (Supplementary Figure S1B, available at https://doi.org/10.1016/j.esmoop.2024.104105). Visualization was carried out using the 'DotPlot' function from 'Seurat'.

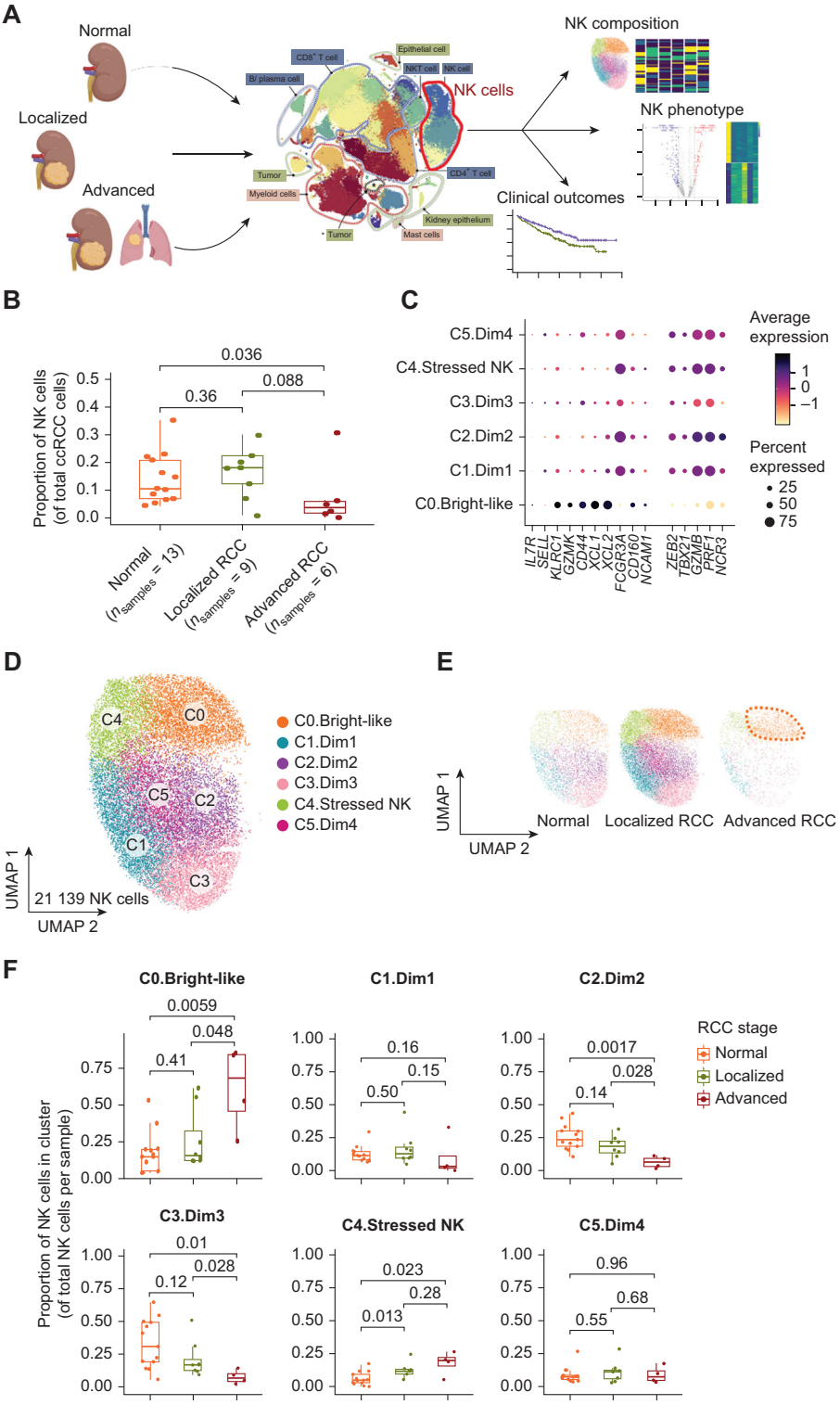


Figure 1. A cluster of non-cytotoxic bright-like NK cells is enriched in advanced ccRCC. (A) Schematic of single-cell transcriptomic profiling of NK cells subsetted from an scRNA-seq dataset of immune cells in advancing disease stages of ccRCC and adjacent normal tissue. (B) Relative proportions of NK cells compared with total immune cell populations (*PTPRC*/CD45+ cell clusters) in normal (non-malignant adjacent) kidney, localized ccRCC tumors (stage I-III), and advanced RCC (stage IV) samples. *P* values displayed for Wilcoxon rank sum test for pair-wise comparison between the different stages of ccRCC (green and red) and control (orange). (C) Dot plot of canonical NK gene markers[14] consisting of markers for bright NK cells and dim NK cells, including annotation of re-clustered NK populations based on the markers. (D) UMAP representation of sub-clustered NK cell populations, displaying substantial heterogeneity (five clusters). (E) UMAP representations of NK cells in normal kidney, localized RCC, and advanced RCC. In metastatic RCC, cluster 0 is highlighted. (F) Box plot of the relative proportion of NK subpopulations (as a proportion of total NK cells), normalized within the sample across disease stages and control with sample count normal = 13, localized = 8, advanced = 4. For box plots, hinges are 25th-75th percentiles; central lines are medians, whiskers are highest and lowest values no greater than 1.5× interquartile range, and dots are outliers.

ccRCC, clear-cell renal cell carcinoma; NK, natural killer; scRNA-seq, single-cell RNA sequencing; UMAP, uniform manifold approximation and projection.

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Analysis of NK cell proportion across ccRCC disease stages and adjacent normal tissue

We defined groups of ccRCC by tissue of origin and stage. Normal stage was defined as samples taken from adjacent, non-malignant, 'normal' tissue (normal annotation in the original immune cell data). Localized ccRCC (T_early and T_localized annotation in the original immune cell data) corresponds to stages I, II, and III of ccRCC (Figure 1A). Advanced ccRCC (T_metast in the original immune cell data) corresponds to stage IV. This definition is used throughout the analysis.

We first calculated the number of NK cells per sample divided by the total number of cells per sample to get the proportion of NK cells per sample (Figure 1B). To determine populations of NK cells enriched in advanced ccRCC, we calculated within-cluster proportions of NK cells normalized by sample. Specifically, for each sample, we calculated the number of NK cells in each cluster and then divided by the total number of NK cells in each sample to get a proportion of NK cells in each cluster relative to total NK cells in each sample. We further labeled each sample with their disease stage.

Next, we carried out two-sided Wilcoxon rank sum test on pair-wise comparisons across the disease stage. An NK cell cluster enriched in advanced ccRCC was determined as the cluster with a significantly higher proportion of NK cells in advanced ccRCC when compared with normal and localized ccRCC, where $P < 0.05$.

All visualizations were done using the 'ggboxplot' and 'stat_compare_means' functions from 'ggpubr v0.6.0' (CRAN; https://cran.r-project.org/package=ggpubr). Stacked barplot of the proportion of NK cells per sample was visualized using 'ggplot2 v3.4.3' (CRAN; https://cran.r-project.org/package=ggplot2).

Differential gene expression analysis

Differential expression testing was carried out using the 'FindMarkers' function from the Seurat package with min.pct = 0.1 while all other parameters were kept as the default setting. Testing was done for each cluster against all other clusters combined. The results were ranked by average $\log_2$ fold change and Bonferroni-adjusted P values (q value). The full list of differentially expressed genes is provided in Supplementary Table S1, available at https://doi.org/10.1016/j.esmoop.2024.104105.

Differentially expressed genes were visualized on a volcano plot (Figure 2A) using 'ggplot2'. Adjusted P values were transformed to a negative $\log_{10}$ scale. Genes were determined as significantly enriched if their average $\log_2$ fold change was >1 and adjusted q value was <0.05. Genes were determined as significantly depleted if its average $\log_2$ fold change was <1 and adjusted q value was <0.05.

Heatmaps of C0.Bright-like marker genes

For the marker heatmaps (Figure 2B and C), the average expression of each gene was calculated from the full dataset using the 'AverageExpression' function from the Seurat package with default parameters. Then, specific marker genes of interest were subsetted from the average expression. Visualization of select genes was done using the 'pheatmap' package with scale = 'row'. For the up-regulated genes, genes were chosen for the top 30 genes with the highest average $\log_2$ fold change. For the down-regulated genes, representative genes from among the top 50 most genes with the lowest $\log_2$ fold change were chosen for visualization (full list available in Supplementary Table S1, available at https://doi.org/10.1016/j.esmoop.2024.104105).

Functional characterization of C0.Bright-like NK cells

We then characterized NK cell gene programs by calculating the gene program scores across the dataset. Cytotoxicity gene sets were defined as *GZMA*, *GZMB*, *GZMK*, *GZMM*, *PRF1*, *LAMP1*, and *LAMP2*. Tissue residency gene sets were defined as *CD69*, *ITGAE*, *CXCR6*, and *ITGA1*.[15] All calculations were done using 'AddModuleScore' from 'Seurat' with default parameters.

For visualizations, we carried out smoothing as previously suggested by best practices for single-cell data analysis.[16] Smoothing for Figure 2D (left) was carried out by taking the arithmetic mean of the module scores over k-nearest neighbors of each cell where k = 634 that is 3% of the total NK cells. For visualization purposes, we plotted the smoothed module scores using the 'VlnPlot' function across the six clusters in Figure 2D (middle) and disease stage (right). We also provided the raw module scores plotted on violin plots in Supplementary Figure S2A and B, available at https://doi.org/10.1016/j.esmoop.2024.104105. Two-sided Wilcoxon rank sum test was carried out on the raw module scores of advanced ccRCC compared with normal and localized ccRCC, respectively.

Signature score calculation and analysis

A set of gene signatures was defined by taking genes with average $\log_2$ fold change of ≥ 1.5 from the differentially expressed genes of C0.Bright-like population resulting in a total of 51 signature genes. Next, we utilized The Cancer Genome Atlas ccRCC cohort (TCGA KIRC) cohort bulk RNA-seq data and its corresponding clinical data from Genomic Data Commons (data downloaded from https://gdc.cancer.gov/about-data/publications/pancanatlas). The clinical data were filtered by retaining samples with corresponding bulk RNA-seq data. We also calculated z-scores from the bulk RNA-seq using the 'scale' function from the R 'base v4.3.1' package (with the following parameters: center = TRUE, scale = TRUE).

Signature score was calculated by taking the arithmetic mean of z-scores for each gene signature across samples. Then, we carried out analysis of the gene signatures in TCGA KIRC cohort along disease progression (Figure 3A). To retain adjacent normal tissue samples from the bulk RNA-seq data, we deduplicated tumor and normal samples taken from the same patient by retaining the samples containing sample barcode '11A' corresponding to a normal sample as defined by TCGA barcode system resulting in 603

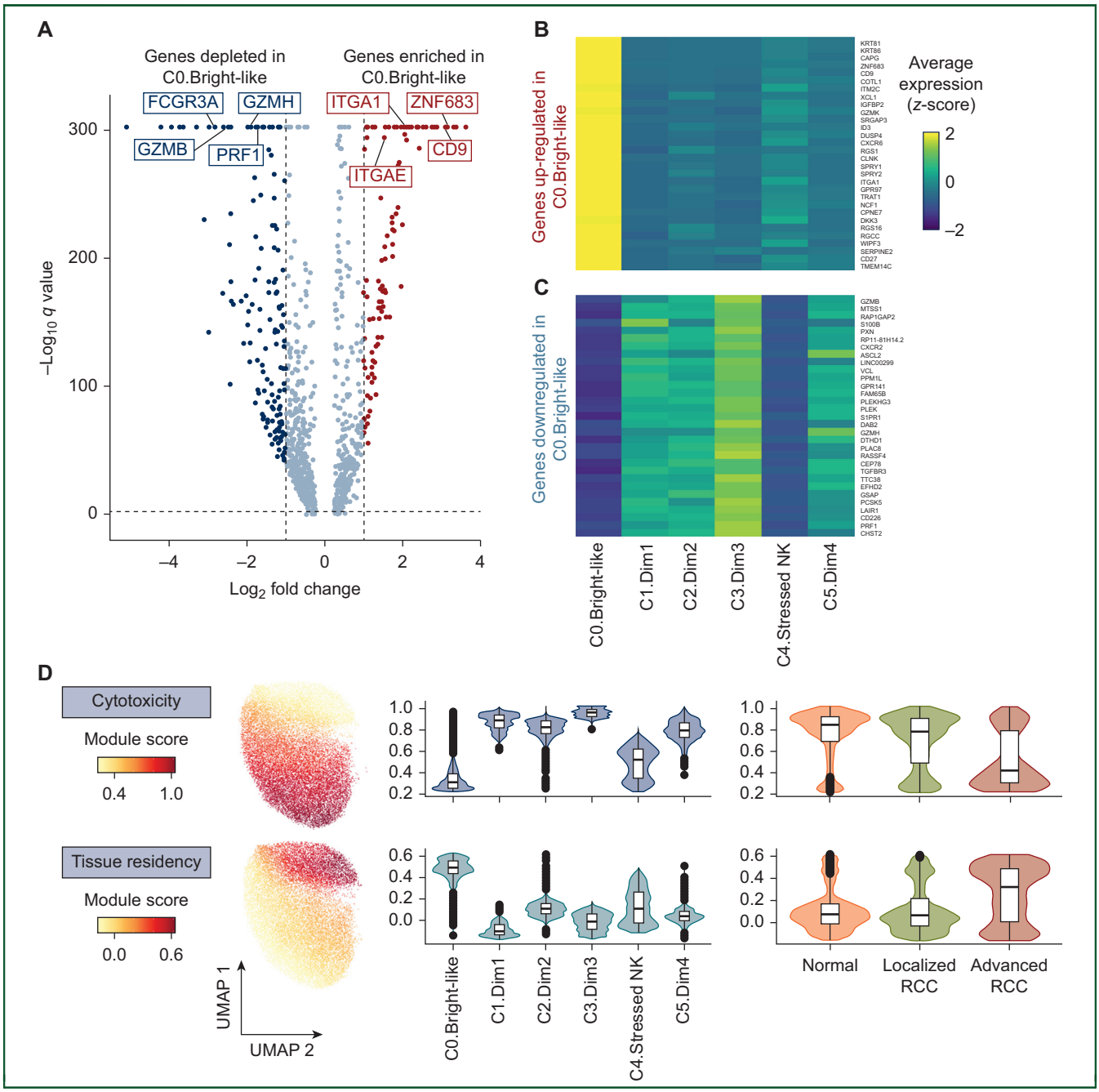


Figure 2. NK cells in advanced ccRCC are depleted of cytotoxic genes and express markers associated with tissue residency. (A) Volcano plot of differentially expressed genes in C0.Bright-like compared with all other NK populations. In red are genes significantly enriched in the C0.Bright-like population (q value < 0.05; $\log_2$ fold change >1; q value is the Bonferroni-corrected P value). In blue are genes significantly depleted in the C0.Bright-like population. Select genes displayed are involved in cytotoxicity and tissue residency. (B and C) Heatmap of selected significantly (B) up-regulated or (C) downregulated in the C0.Bright-like cluster compared with all other NK populations. (D) Characteristics of the C0.Bright-like NK population. (Left) UMAP feature plots show module scores of defined biologically relevant gene sets smoothed over 634 neighbor cells (see 'Materials and methods' section). Violin plots for smoothed module scores are shown in all NK subpopulations (middle) and across RCC stages (right).
ccRCC, clear-cell renal cell carcinoma; NK, natural killer; RCC, renal cell carcinoma; UMAP, uniform manifold approximation and projection.

RNA-seq samples with signature scores. For visualization, results were plotted using 'ggboxplot'. Lastly, we carried out pair-wise two-sided Wilcoxon rank sum tests for C0.Bright-like signature scores across ccRCC progression where $P < 0.05$.

Further, we investigated any differences in the C0.Bright-like signature scores between male and female patients in the TCGA KIRC (Supplementary Figure S3C, available at https://doi.org/10.1016/j.esmoop.2024.104105) and CheckMate 025 (Supplementary Figure S3D, available at https://doi.org/10.1016/j.esmoop.2024.104105) cohorts. Finally, we compared C0.Bright-like signature scores between sarcomatoid/rhabdoid and non-sarcomatoid/rhabdoid patients in the CheckMate 025 cohort (Supplementary Figure S3E, available at https://doi.org/10.1016/j.esmoop.2024.104105), using a two-sided Wilcoxon

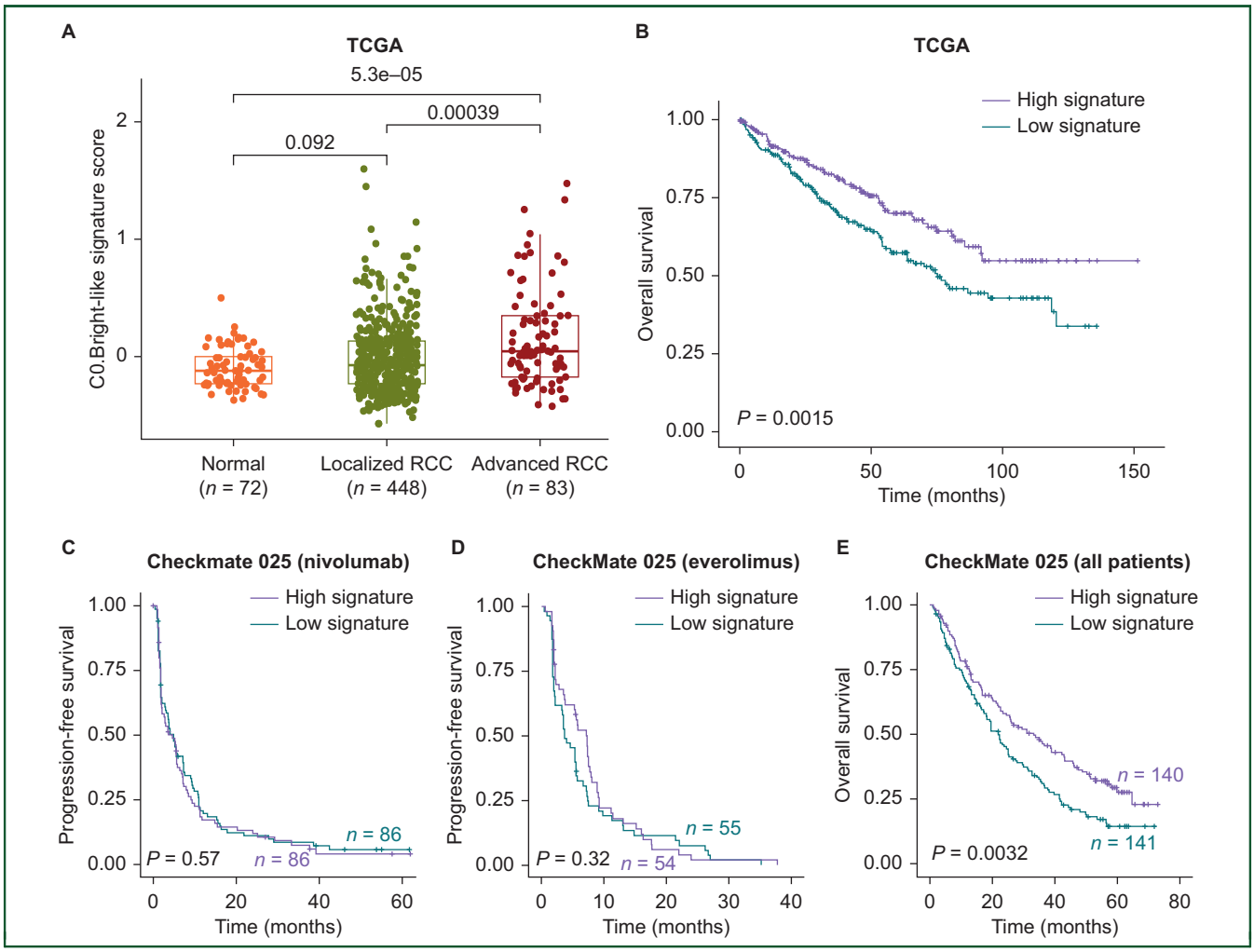


Figure 3. C0.Bright-like NK cells are associated with worse overall survival in external clinical datasets. (A) Box plots showing significantly higher signature score of C0.Bright-like population in advanced RCC compared with normal and localized RCC in the external The Cancer Genome Atlas ccRCC cohort (TCGA KIRC; two-sided Wilcoxon rank sum test for pair-wise comparison). (B) Overall survival for TCGA KIRC cohort based on high gene signature for C0.Bright-like (≥median) versus low signature expression (high versus low threshold at the median; log-rank test). Higher signature for the C0 cluster indicates a lower survival. (C) PFS for CheckMate 025 cohort patients within the nivolumab treatment group based on high gene signature for C0.Bright-like (≥median) versus low signature expression, showing no significant difference in PFS. Log-rank test for significance. (D) PFS for CheckMate 025 cohort patients within the everolimus treatment group based on high gene signature for C0.Bright-like (≥median) versus low signature expression, showing no significant difference in PFS. Log-rank test for significance. (E) Overall survival for all CheckMate 025 cohort patients based on high gene signature for C0.Bright-like (≥median) versus low signature expression, showing worse overall survival for patients with a high signature score. Log-rank test for significance.
PFS, progression-free survival; RCC, renal cell carcinoma; TCGA, The Cancer Genome Atlas; UMAP, uniform manifold approximation and projection.

rank sum test for significance (at a significance threshold of <0.05). The designations for sex and tumor pathology (sarcomatoid or rhabdoid status) are from the respective cohort metadata.

Survival analysis in TCGA and CheckMate 025 cohorts

We calculated OS and PFS on TCGA KIRC and CheckMate 025 cohorts (NCT01668784) using Kaplan–Meier analysis.[17] High signature group was defined as having signature scores greater than or equal to the median versus the low signature group having scores less than the median.

For the TCGA KIRC cohort, we retained the tumor sample from each patient. For the CheckMate 025 cohort, we looked at patients receiving nivolumab, everolimus, and all patients. We determined differences in OS and PFS between high signature and low signature groups using two-sided log-rank test at a significance level of 0.05. All analyses were carried out using the 'survfit' function from 'survival v3.5.5' (CRAN; https://cran.r-project.org/package=survival) and 'survminer v0.4.9' (CRAN; https://cran.r-project.org/package=survminer). Visualizations were done using the 'ggsurvplot' function.

Functional analysis in vitro

To determine whether our findings were functionally relevant *in vitro*, we collected tumor samples from an additional five patients with ccRCC. Along with tumor collection, peripheral blood samples were collected as well as adjacent healthy tissue in cases where this was available (Supplementary Tables S2-S4, available at https://doi.org/10.1016/j.esmoop.2024.104105).

Dissociation of tumor tissue and healthy tissue was carried out as previously described[3] and single cells were frozen in Bambanker (FUJIFILM Wako Chemicals, Richmond, VA) at −80°C. Peripheral blood mononuclear cells (PBMCs) were extracted from peripheral blood using Ficoll gradient and EasySep tubes (StemCell, Vancouver, Canada). PBMCs were frozen in Bambanker at −80°C.

For the functional analysis, samples were thawed in RP10 with interleukin 15 (IL-15) 1 ng/ml and rested overnight in an appropriate volume. The following day tissue or tumor samples were sorted as live (using violet dead viability dye) and CD45+ to select lymphocytes, or CD56+ to select NK cells. The cells were then rested overnight in RP10 and IL-15 (1 ng/ml) and used the next day in an *in vitro* assay. To set up the flow-based functional assay, cells were counted and plated out in duplicate as NK alone and NK at a 1 : 1 ratio with K562. In three cases, there were sufficient cells to plate out a third condition with cytokine-induced memory-like (CIML) cytokines IL-15 (50 ng/ml), IL-18 (50 ng/ml), and IL-12 (20 ng/ml) to test if cytokine activation could rescue the functionality of NK cells from the tumor. K562 target cells were stained before plating out with CellTraceViolet dye (Thermo Fisher Scientific, Waltham, MA,) for 20 min at 37°C and washed once in RP10 before counting.

After 1 h co-culture, GolgiPlug and GolgiStop (BD Biosciences, Franklin Lakes, NJ) were added to each well and the cells were incubated for a further 5 h before staining. Cells were stained for CD56, CD3, CD49a, CD9, CD107a, and interferon-γ (IFN-γ) and run on a BD Fortessa flow cytometer. Data were analyzed using FlowJo (BD Biosciences, Franklin Lakes, NJ).

RESULTS

Single-cell RNA sequencing identified a unique dysfunctional NK cell cluster enriched in the tumor microenvironment of advanced ccRCC

Single-cell transcriptomic data from tissue samples including tumor and adjacent normal tissue were obtained and analyzed from 13 patients with RCC, all of whom had received no prior systemic therapy for RCC.[3] The median patient age was 64.5 years (range 50-83 years). Five patients (38%) had stage IV disease at the time of sample collection. NK cells were identified using classical lineage markers (*NCAM1*/CD56, *NCR1*/NKp46). In total, we identified and carried out scRNA-seq on 21 139 individual NK cells, out of a total cell population of 164 722 cells (Figure 1A). Among all immune cells, the proportion of NK cells was lower in tumor tissue from patients with advanced/metastatic RCC compared with normal tissues (Figure 1B, Supplementary Figure S1A, available at https://doi.org/10.1016/j.esmoop.2024.104105).

Unsupervised graph-based clustering analysis was carried out to identify distinct NK cell populations, revealing six distinct NK cell subsets (Figure 1C and D). The majority of the NK cell subsets expressed genes classically associated with cytotoxic NK cells including *GRZB, PRF1*, and *NCR1* (Figure 1C) and all of the clusters had a high expression of *KLRF1* and a low expression of *KLRF2,* confirming their NK cell phenotype (Supplementary Figure S1B, available at https://doi.org/10.1016/j.esmoop.2024.104105).

Although the proportion of NK cells decreased in advanced ccRCC, there was a clear enrichment of a specific population of NK cells expressing genes associated with CD56^{bright} NK cells (*GZMK, XCL1, XCL2*) in patients with advanced disease (Figure 1E and F, Supplementary Figure S1C, available at https://doi.org/10.1016/j.esmoop.2024.104105). Advanced disease stages were also enriched for a cluster of NK cells expressing genes associated with stress (Figure 1F).

We further characterized differences among NK cell subpopulations, with particular attention to the enriched C0.Bright-like population. A volcano plot of differentially expressed genes in C0.Bright-like cells compared with all other NK cells is shown in Figure 2A. Notable genes expressed highly in the bright-like cluster were *CD9*, *ITGA1*/CD49a, *ITGAE*/CD103, and *ZNF683*, while there was decreased expression of the CD16/*FCGR3A* receptor, *GZMB*, *GZMH*, and *PRF1* expression. This was consistent with decreased cytotoxicity and a 'tissue-resident' NK cell phenotype (Supplementary Figure S2, available at https://doi.org/10.1016/j.esmoop.2024.104105).

Next, we identified a gene expression signature (GES) using differentially expressed genes in the C0.Bright-like cluster compared with other NK cells. This transcriptomic signature effectively distinguished cluster 0 cells from other cell clusters (Figure 2B and C). Using GESs for cellular cytotoxicity and tissue residency, we confirmed that the C0.Bright-like NK cells had a tissue residency phenotype with diminished cytotoxicity (Figure 2D, Supplementary Figure S2, available at https://doi.org/10.1016/j.esmoop.2024.104105).

Bright-like NK cells are associated with worse clinical outcomes in ccRCC

We investigated whether the GES corresponding to C0.Bright-like NK cells is associated with advanced disease stage and worse clinical outcomes in external datasets. Using TCGA KIRC transcriptomic dataset, we found that the C0.Bright-like signature was enriched in patients with advanced/metastatic RCC compared with normal subjects or those with localized RCC (Figure 3A). Furthermore, we dichotomized patients within the TCGA dataset based on high versus low expression of the C0.Bright-like signature (based on median GES expression). Patients with higher baseline signature expression had worse OS compared with those with low baseline expression (log-rank $P = 0.0029$, Figure 3B). Similarly, when applying this signature to transcriptomic data from patients with metastatic ccRCC who participated in the CheckMate 025 clinical trial, patients with higher signature expression had worse OS (log-rank $P = 0.003$, Figure 3E).

The C0.Bright-like transcriptomic signature was prognostic but not predictive, as evidenced by the lack of a difference in PFS among patients with high versus low signature expression in either the nivolumab or everolimus treatment arms (Figure 3C and D). Although the C0.Bright-like signature was not associated with radiographic response to nivolumab or

everolimus (Supplementary Figure S3A and B, available at https://doi.org/10.1016/j.esmoop.2024.104105), it was enriched in male patients, and among patients with tumors that had sarcomatoid/rhabdoid features, both of which are patient populations known to be associated with worse prognosis in ccRCC (Supplementary Figure S3C-E, available at https://doi.org/10.1016/j.esmoop.2024.104105).[18,19] We also investigated the association of the C0.Bright-like transcriptomic signature with OS in the nivolumab and everolimus treatment arms of CheckMate 025 (Supplementary Figure S3A and B, available at https://doi.org/10.1016/j.esmoop.2024.104105), and observed that the patients in the nivolumab arm had significantly lower OS when they had a higher baseline C0.Bright-like signature ($P = 0.035$). However, no significant association with OS was observed in the everolimus arm ($P = 0.19$).

Tumor-resident NK cells isolated from patients with ccRCC are dysfunctional

We next evaluated whether NK cells isolated from primary ccRCC tumors and metastatic sites have impaired functional activity *in vitro*. Live, CD45+ immune cells were sorted from ccRCC tumors from three patients and the metastatic site from two patients with ccRCC (Supplementary Table S2, available at https://doi.org/10.1016/j.esmoop.2024.104105) and stained the subsequent day for CD56, CD3, CD49a, and CD9 to identify our subsets of interest (based upon the expression of some of the most highly expressed genes in cluster 0—CD49a/*ITGA1* and *CD9*) (Figure 4A and B). We identified distinct populations of intratumoral NK cells, of which the largest proportion was CD49a−CD9− NK cells followed by CD49a+CD9− NK cells and CD49a+CD9+ NK cells (Figure 4B and C). After a 6-h co-culture with K562 target cells or stimulation with cytokines (IL-15, IL-12, and IL-18), as predicted from the single-cell transcriptomic data, we observed no differences in effector cytokine production (the proportion of IFN-γ-positive cells) when comparing CD49a+CD9+ NK cells versus CD49a−CD9− NK cells (Figure 4D). However, cytotoxicity (as measured by the surrogate degranulation marker CD107) was lower among CD49a+CD9+ NK cells and higher among CD49a−CD9− NK cells (Figure 4E). Further, cytokine stimulation of tumor-derived CD49a+CD9+ NK cells did not restore their cytotoxic activity (Supplementary Figure S4, available at https://doi.org/10.1016/j.esmoop.2024.104105). This suggests that within ccRCC tumors, tissue-resident CD49a+CD9+ NK cells retain cytokine production but have impaired degranulation/cytotoxic function.

For two patients, we isolated NK cells from the adjacent non-malignant tissue and sorted based on CD45+ and stained as described above. In the healthy tissue, there were very few CD49a+ or CD9+ NK cells, in line with our transcriptomic data. The CD49a+CD9+ NK cells isolated from normal kidney tissue, however, showed strong IFN-γ production and degranulation when stimulated with K562 target cells (Supplementary Figure S5, available at https://doi.org/10.1016/j.esmoop.2024.104105).

We also evaluated PBMCs from four of the patients, where we observed a relatively high percentage of CD49a+ NK cells, but a small percentage of CD49a+CD9+ cells (Supplementary Figure S6A and B, available at https://doi.org/10.1016/j.esmoop.2024.104105). In contrast to tumor-infiltrating NK cells, we found no differences in cytokine production or cytotoxic function among CD49a+CD9+ versus CD49a−CD9− NK cells, with in some cases CD49a+CD9+ cells showing higher functionality. This suggests that NK cells acquire a 'tissue-resident' phenotype with impaired cytotoxicity within the tumor microenvironment and not in the peripheral circulation (Supplementary Figure S6C and D, available at https://doi.org/10.1016/j.esmoop.2024.104105).

DISCUSSION

In this study, we evaluated NK cell gene expression and function among patients with ccRCC. We showed that disease progression is associated with diminished infiltration of cytotoxic NK cells and enrichment of non-cytotoxic, bright-like NK cells within the tumor microenvironment. These bright-like NK cells have a tissue-resident phenotype and express genes commonly associated with non-cytotoxic $CD56^{bright}$ NK cells. A GES based on this enriched NK cell cluster was associated with worse OS in two large, independent cohorts. In functional studies, tissue-resident NK cells from RCC tumors showed decreased cytotoxicity/degranulation potential when co-cultured with tumor target cells or upon cytokine stimulation. In contrast, NK cells from adjacent healthy normal kidney tissue and the peripheral blood of these patients maintained full function.

These data add to our prior understanding of dysfunctional immune circuits in ccRCC. In prior single-cell RNA and T-cell receptor sequencing studies, we found that metastatic ccRCC tumors have a complex microenvironment of immune cells characterized by multiple axes of immune dysfunction including decreased CD4+ T cells, increased abundance of terminally exhausted CD8+ T cells, decrease in pro-inflammatory macrophages, and increase in immunosuppressive, C1q+ APOE+ TREM2+ tumor-associated macrophages.[3] Together, our findings suggest that the clinical progression of ccRCC is accompanied by NK cell dysfunction, and that ccRCC thereby evades detection and elimination by cytotoxic immune cells.

Prior studies have described the presence of ILC1/tissue-resident NK cells within human tumors including ccRCC.[10,20] Compared with previous work, we show that NK cell subsets differ among early versus late clinical stages in ccRCC. In contrast to the prior work by Kansler and colleagues, we show that these CD49a+ NK cells can be detected in both tumor and blood.[10] Our findings suggest that CD49a+ NK cells retain cytotoxic function when in circulation but are dysfunctional within the tumor microenvironment.

The tissue-resident NK cells we identified from ccRCC differ from those previously described in chromophobe RCC. In the current study, cells had a poor cytotoxicity/

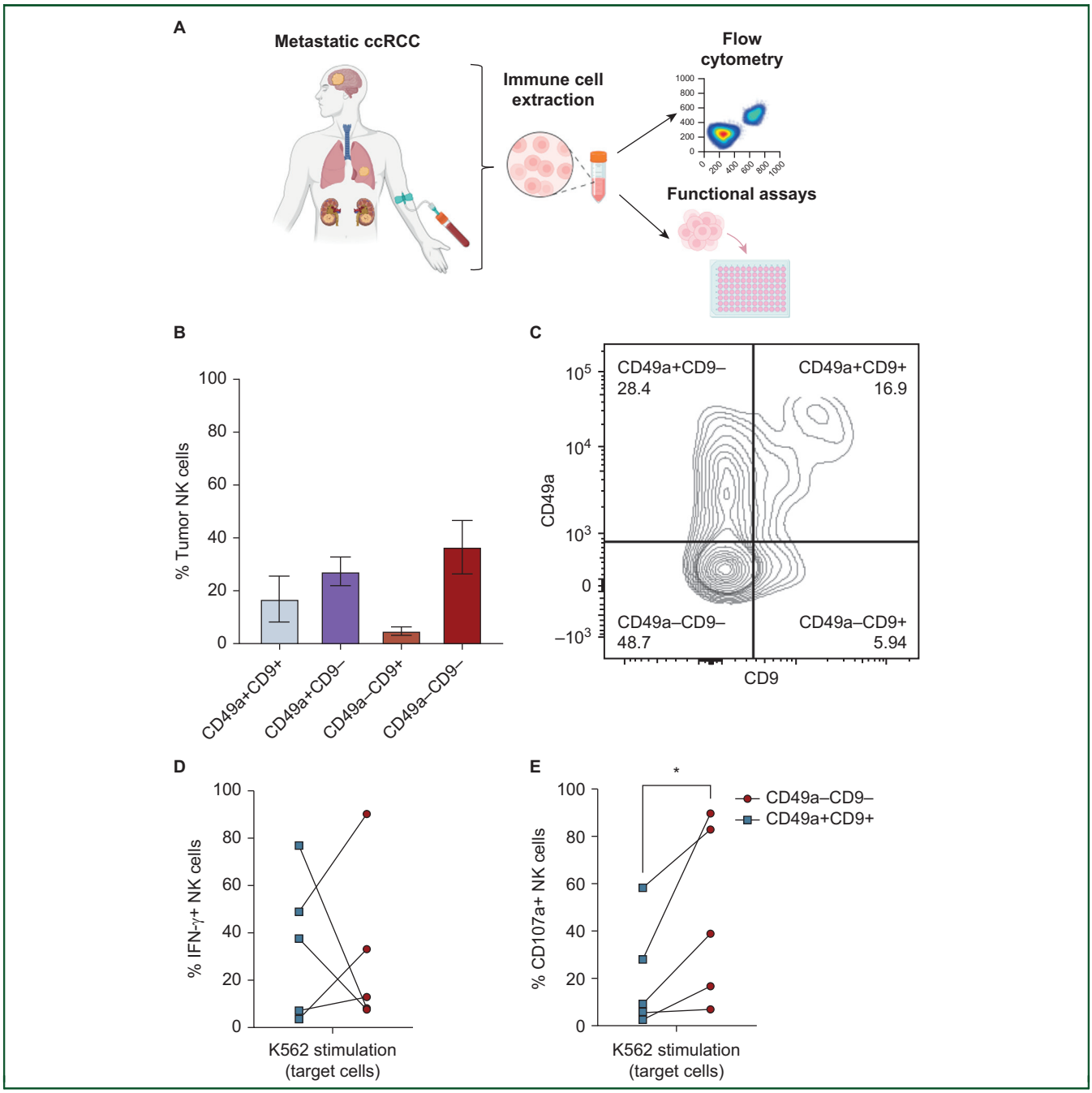


Figure 4. CD49a+CD9+ NK cells isolated from metastatic tumors lack antitumor functional activity. (A) Schematic describing *in vitro* assays to assess the function of NK cells isolated from ccRCC patients with metastatic disease. (B) Proportions of NK cells isolated from the tumor (sorted live, CD56+ cells) based on CD49a and CD9 expression. (C) Flow cytometry gating strategy. (D) Graph showing the percentage of CD49a+CD9+ and CD49a−CD9− NK cells that are IFN-γ positive after 6-h co-culture with K562 target cells. (E) Graph showing the percentage of CD49a+CD9+ or CD49a−CD9− NK cells that are CD107a+ after co-culture for 6 h with K562 target cells. $n = 5$ renal cell carcinoma cases. *$P < 0.05$ one-tailed paired *t*-test.
ccRCC, clear-cell renal cell carcinoma; IFN-γ, interferon-γ; NK, natural killer.

degranulation response to cytokine stimulation with IL-15, IL-12, and IL-18, but were able to produce IFN-γ.[10] Our data therefore suggest that exogenous cytokine stimulation alone may be inadequate to restore the antitumoral NK cell response in patients with metastatic ccRCC. Nonetheless, it raises the possibility that other techniques to improve NK cell antitumor toxicity could have therapeutic implications for ccRCC. Potentially promising approaches could include the use of cytokine stimulation on peripheral blood NK cells to generate CIML NK cells, an approach that has been previously described and which has shown activity in both hematologic and solid tumors.[21,22] This is supported by recent work demonstrating that exogenous IL-15 can improve cytotoxic activity of tissue-resident intratumoral NK cells and improve tumor control.[23] Other potential approaches to enhance NK cell potency and persistence could include the use of engineered NK cells or engager molecules, or concurrent checkpoint inhibition to disrupt inhibitory circuits.[24,25]

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Our study is subject to several limitations. While our data represent a wide range of early versus late disease stages in ccRCC, the majority of our samples are from nephrectomy specimens and a greater number of metastatic samples would be needed to explore whether NK cell subpopulations differ across metastatic sites. Our small sample size in the validation studies is also a limitation. In future studies we would endeavor to expand this analysis to include a higher number of metastatic and peripheral samples. While we identified phenotypically similar populations of CD49a+/CD9+ NK cells in the peripheral compartment, molecular barcoding studies would be needed to specifically verify whether these cells represent the same underlying population as are found in the tumor. Future studies could include an analysis of spatial relationships among NK cells and other immune cells in the tumor microenvironment, which could be better elucidated through spatial transcriptomics.[26,27] A further limitation is our use of a subsequent line (second and third) dataset for correlations with overall response and PFS. Future studies will validate our findings in larger first-line immunotherapy-based studies.

In summary, among patients with ccRCC, a retrospective, single-cell transcriptomic analysis revealed heterogeneous NK cell populations. A dysfunctional resident NK cell phenotype is enriched among patients with metastatic disease and is associated with worse survival in patients with advanced ccRCC. Restoration of NK cell function could be a future therapeutic opportunity among patients with ccRCC.

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